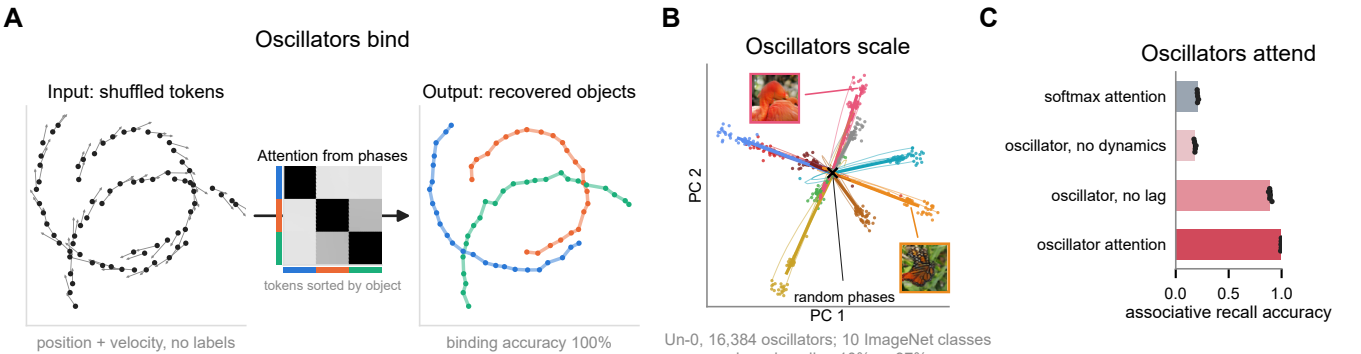

A
Oscillators bind
Input: shuffled tokens
Attention from phases
tokens sorted by object
Output: recovered objects
position + velocity, no labels
binding accuracy 100%
B
Oscillators scale
PC 2
PC 1
random phases
Un-0, 16,384 oscillators; 10 ImageNet classes
class decoding 10% → 97%
C
Oscillators attend
softmax attention
oscillator, no dynamics
oscillator, no lag
oscillator attention
0.0
0.5
1.0
associative recall accuracy
1 layer each, ~58k parameters
chance 3%